(a) Weight Packer offline

Filter Weights W

	A	B
k0	2	0
k1	0	3
k2	0	1
k3	1	0

row masks

1	0	0
0	1	0
0	1	0
1	0	0

∧

0 No row used twice → share

Packed Column (weight | filter label)

k0	2	A
k1	3	B
k2	1	B
k3	1	A

8 PEs → 4 PEs: every PE holds a real weight

≤ F = 4 filters per column

Weight Memory

packed offline, per (filter block, K-tile)
column = 4 × (8-b weight + 2-b label) + ids

4 Load weights → PEs

(b) Pixel Packer run time

Spike Buffer S[k][p] (T = 4 bits t0..t3)

	p0	p1	p2	p3
k0	1 0 0 0	0 0 0 0	0 1 0 0	0 0 1 1
k1	0 0 0 0	0 0 1 0	0 0 1 1	0 0 0 0
k2	0 1 0 0	0 0 1 1	0 0 0 0	1 0 0 0
k3	0 0 1 1	0 0 0 0	1 0 0 0	0 1 0 0

1 Read a K-tile

First-Fit (≤ 4 open groups, ≤ G = 4 pixels)

p1	G1 = {p0}	∧ = 0	join G1
p2	G1 = {p0,p1}	≠ 0 @ (k1,t2)	open G2
p3	G2 = {p2}	∧ = 0	join G2

2 Check: no shared (row, t) bit

Group Buffer (merged bits + 2-b pixel labels)

G1 = p0 + p1

k0	1	0	0	0
k1	0	0	1	0
k2	0	1	1	1
k3	0	0	1	1

G2 = p2 + p3

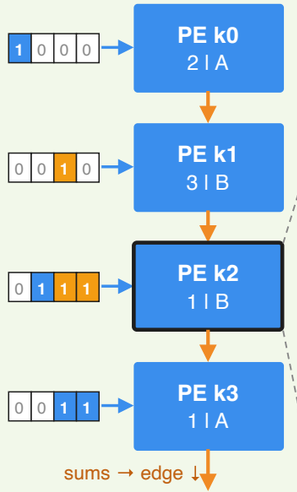
k0	0	1	1	1
k1	0	0	1	1
k2	1	0	0	0
k3	1	1	0	0

3 Merge: 4 pixels → 2 groups

8 lanes, one K-tile each

(c) Weight-Stationary PE Array

5 Stream G1 into the rows



PE k2 · weight 1 | B · G1 arrives

	0	1	1	1
	–	p0	p1	p1
w=1	+	+	+	+
	t0	t1	t2	t3

6

running sums (filter, pixel) × t

A,p0	2	0	0	0
A,p1	0	0	0	0
B,p0	0	1	0	0
B,p1	0	0	4	1

adder t adds w if bit t = 1; the filter label (B) and the bit's pixel label pick the sum — no search, no compare

4 × 4 PEs, T adders per PE: all timesteps of a train in one cycle.
Spikes move right, running sums move down; weights stay (shadow registers load the next pass meanwhile).

(d) Edge Unit & LIF

Edge FIFO (touched sums only)

(B,p1)	0	0	4	1
(B,p0)	0	1	0	0
(A,p0)	2	0	1	1

Addr (filter, pixel) 4 RMW / col

7 Enqueue 8 RMW (+=)

Output Buffer (sums across K-tiles)

	p0	p1	p2	p3
A	2 0 1 1	0 0 0 0	1 2 0 0	0 1 2 2
B	0 1 0 0	0 0 4 1	0 0 3 3	1 0 0 0

LIF Units ×16 (after the last K-tile)

$V \leftarrow V/2 + I_t$; spike if $V \geq \theta$, then $V -= \theta$

9 Fire → next layer's spike trains

	p0	p1	p2	p3
A	1 0 0 0	0 0 0 0	0 1 0 0	0 0 1 1
B	0 0 0 0	0 0 1 1	0 0 1 1	0 0 0 0

Exactly the dense GEMM + LIF result:
packing is lossless; labels route every add.